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Homes Experiences

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Who
Add guests

Airbnb Listings EDA Project: New York 2025

Stay near Mahalakshmi Race Course

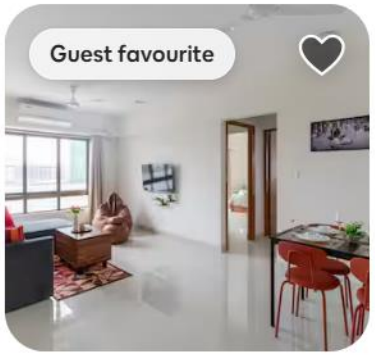
Apartment in Mahalakshmi
₹10,839 for 1 night · ★ 4.89

Flat in Santacruz East
₹7,658 for 1 night · ★ 4.91

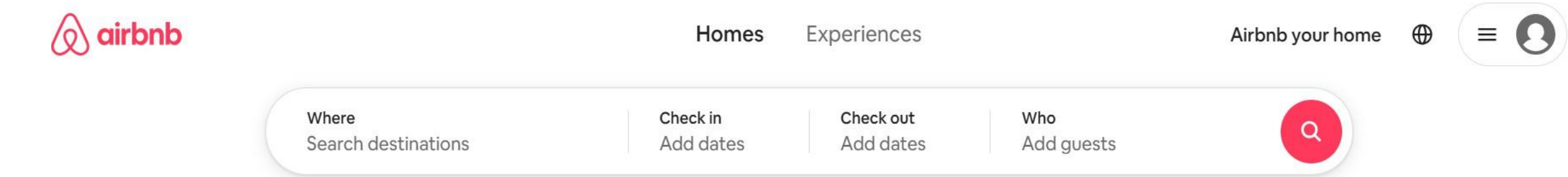
Flat in Bandra East
₹8,559 for 1 night · ★ 5.0

Flat in Bandra West
₹7,733 for 1 night · ★ 4.82

Flat in Bandra East
₹9,130 for 1 night · ★ 5.0

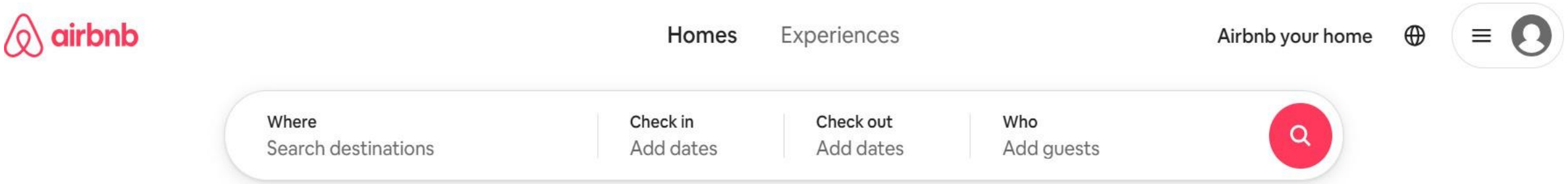


Flat in Santacruz East
₹8,558 for 1 night · ★ 4.94



Project Overview

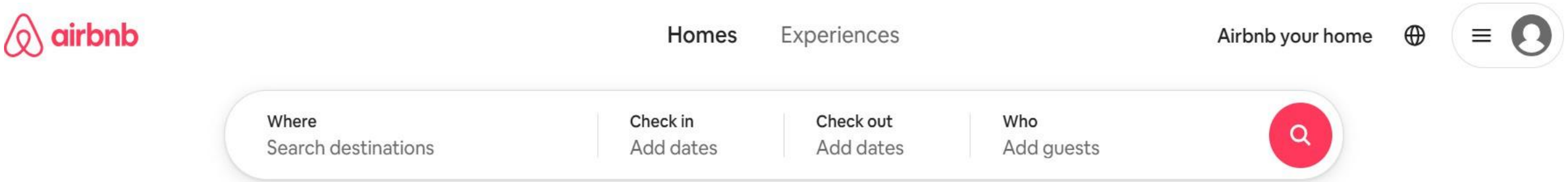
This project performs Exploratory Data Analysis (EDA) on New York Airbnb data to uncover trends and patterns in rental listings. Using libraries like Pandas, Numpy, Matplotlib, Seaborn for cleaning, visualization and analysis.



Objective

The goal of this project is to:

- Analyze room types, prices, and availability across different neighborhoods.
- Understand host behavior and listing patterns.
- Detect potential outliers in prices.
- Provide recommendations for guests and hosts based on insights.



Dataset Overview

- Data source – Kaggle
- Number of rows and columns – 20770 X 22
- Key variables / Features - Latitude, Longitude, Price, Minimum nights, Number of reviews, Reviews per month, Availability 365, Beds, Neighbourhood group

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Import libraries

```
[ ]  import pandas as pd
     import numpy as np
     import matplotlib.pyplot as plt
     import seaborn as sns
```

Loading dataset

```
df = pd.read_csv("AB_NYC_2019.csv")
```

```
df
```

Data exploration

```
df.head()
```

↓

```
df.tail()
```

↓

```
df.shape
```

↓

```
df.info()
```

↓

```
df.describe()
```

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Data cleaning

```
df.isnull().sum()

#dropping the missing value rows
df.dropna(inplace=True)

df.isnull().sum()
```

	0
id	0
name	0
host_id	0
host_name	0
neighbourhood_group	0
neighbourhood	0

```
# Handling duplicate datas
df.duplicated().sum()

#dropping duplicate records
df.drop_duplicates(inplace=True)
df.duplicated().sum()
```

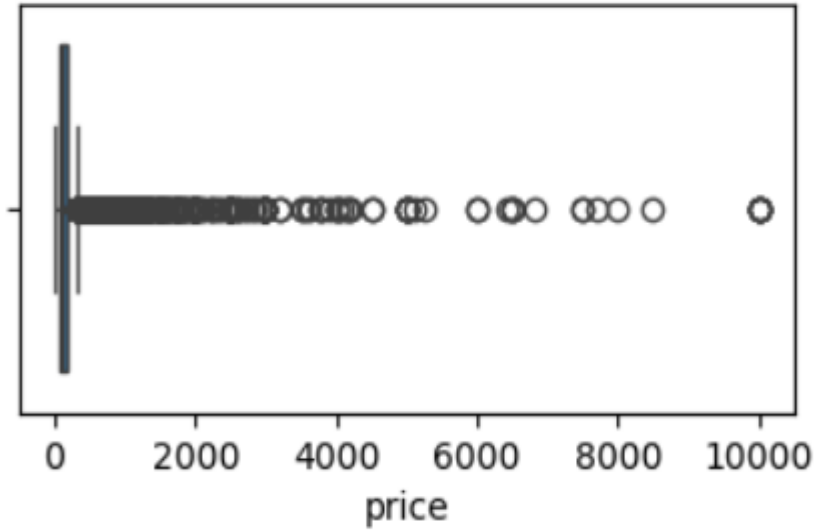
```
np.int64(0)
```

```
# Type casting
df.dtypes

df['id']=df['id'].astype(object)
df.dtypes
df['host_id']=df['host_id'].astype(object)
df.dtypes
```

EDA – Univariate Analysis

Checking Outliers for price column

```
# EDA - Univariate Analysis
# Checking outliers for price column
plt.figure(figsize=(4,2))
sns.boxplot(data = df,x='price')
plt.show()
```



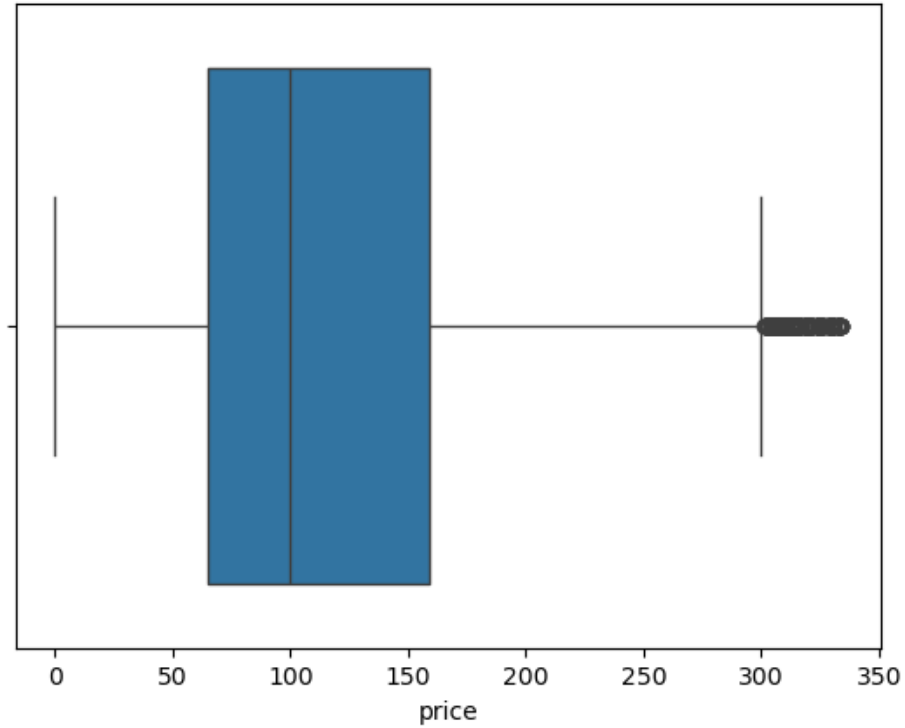

```
# Remove outliers using IQR
Q1 = df['price'].quantile(0.25)
Q3 = df['price'].quantile(0.75)
IQR = Q3 - Q1

df = df[(df['price'] >= Q1 - 1.5*IQR) & (df['price'] <= Q3 + 1.5*IQR)]
df
```

EDA – Univariate Analysis

After removing outliers using IQR

```
# Remove outliers using IQR
Q1 = df['price'].quantile(0.25)
Q3 = df['price'].quantile(0.75)
IQR = Q3 - Q1

df = df[(df['price'] >= Q1 - 1.5*IQR) & (df['price'] <= Q3 + 1.5*IQR)]
df
```

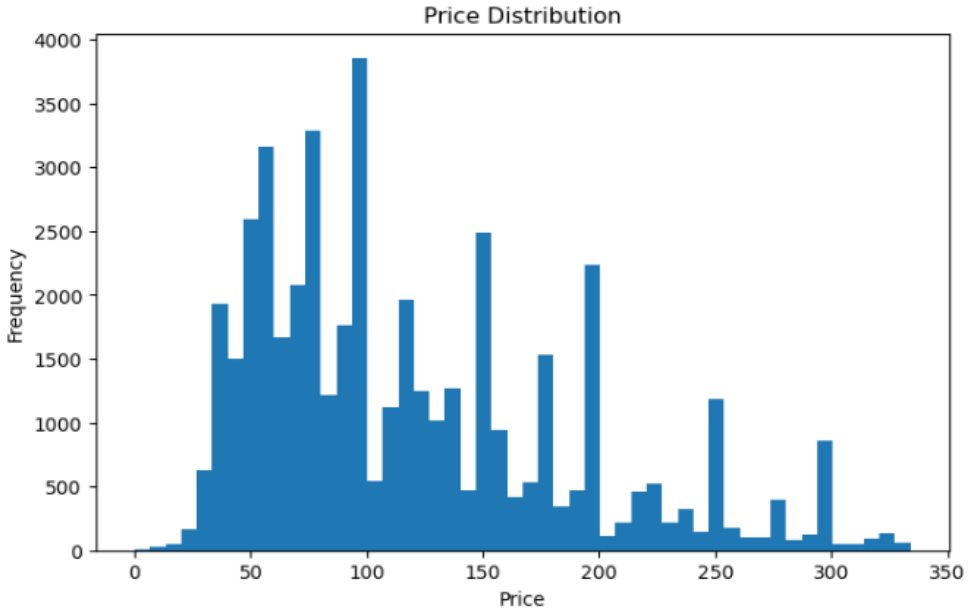
```
sns.boxplot(x=df['price'])
plt.show()
```



EDA – Univariate Analysis

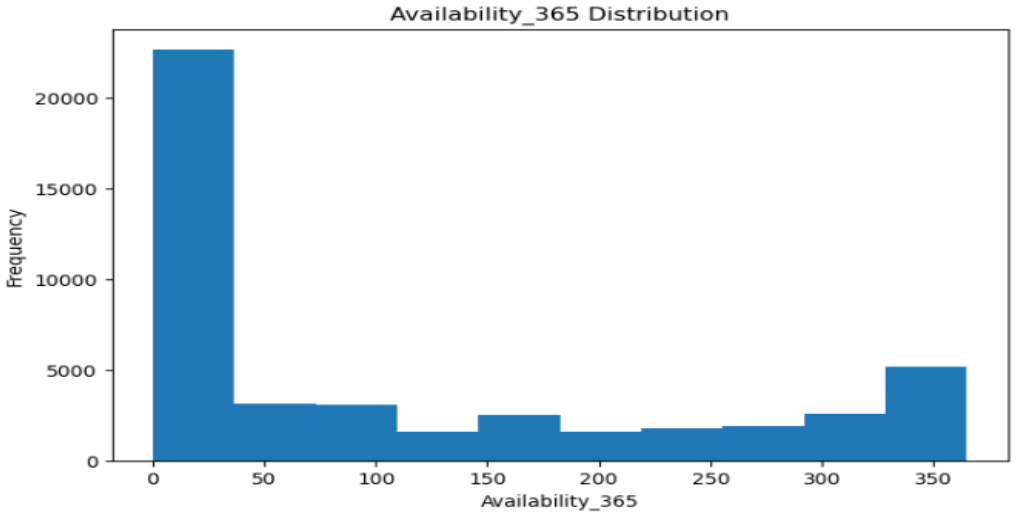
Data distribution of price column

```
# Histogram
plt.figure(figsize=(8,5))
plt.hist(df['price'], bins=50)

plt.title("Price Distribution")
plt.xlabel("Price")
plt.ylabel("Frequency")

plt.show()
```



Data distribution of Availability 365 column

```
# Data Distribution of Availability 365 column
plt.figure(figsize=(8,5))
plt.hist(df['availability_365'])

plt.title("Availability_365 Distribution")
plt.xlabel("Availability_365")
plt.ylabel("Frequency")
plt.show()
```



EDA – Univariate Analysis

Data summary of neighbourhood group with price

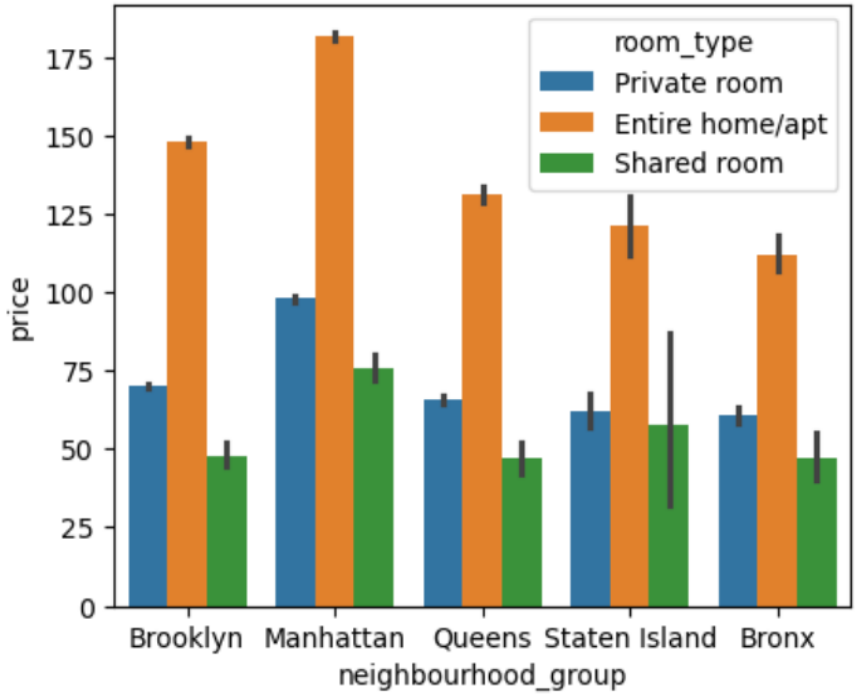
```
# Data Summary of Neighborhood Group with Price
df.groupby('neighbourhood_group')['price'].mean()
```

```
neighbourhood_group
Bronx             77.365421
Brooklyn         105.699614
Manhattan        145.952835
Queens            88.904437
Staten Island     89.235616
Name: price, dtype: float64
```

EDA – Bivariate Analysis

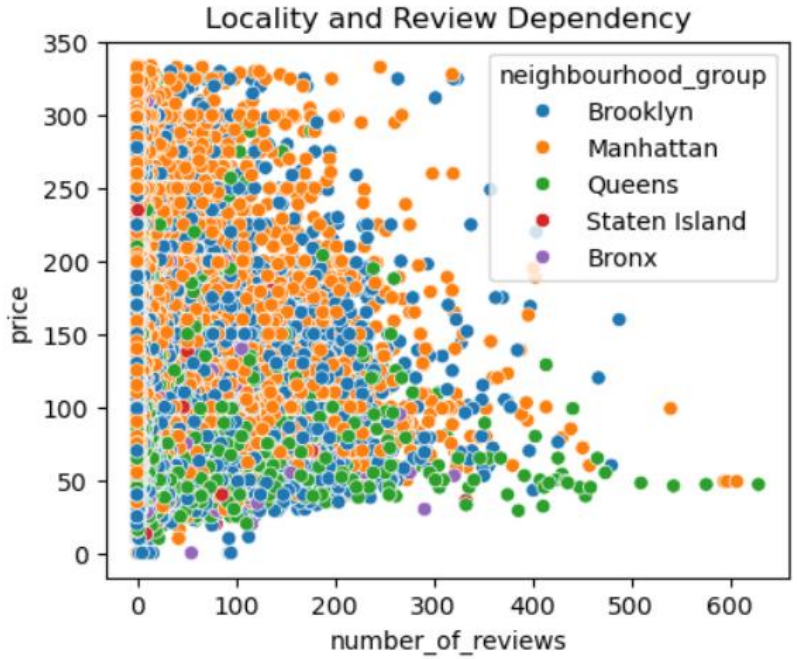
Price dependency on room type

```
# EDA - Bivariate Analysis
plt.figure(figsize=(5,4))
sns.barplot(data = df,x='neighbourhood_group',y='price',hue = 'room_type')
plt.show()
```



Price relationship with number of reviews

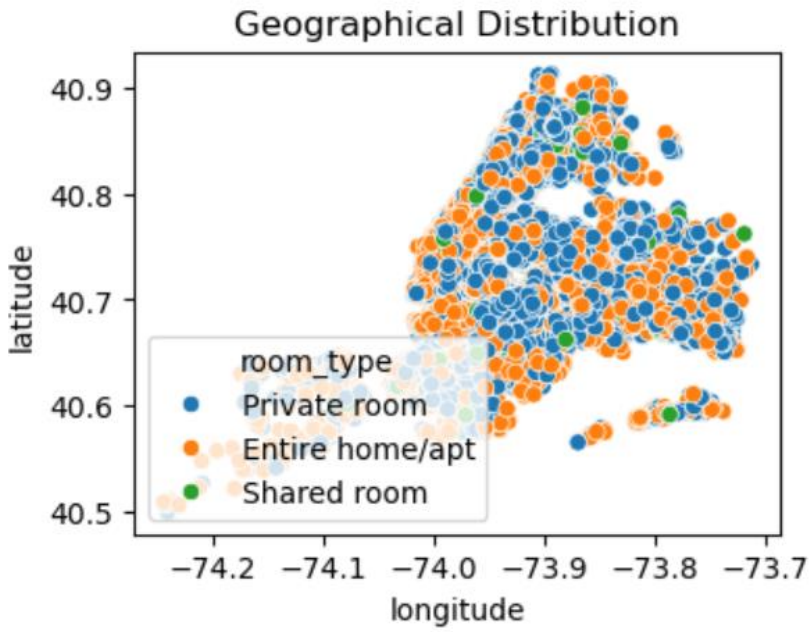
```
# Price Relationship with Number of Reviews?
import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(5,4))
sns.scatterplot(data = df,x='number_of_reviews',y='price',hue = 'neighbourhood_group')
plt.title('Locality and Review Dependency')
plt.show()
```



EDA – Bivariate Analysis

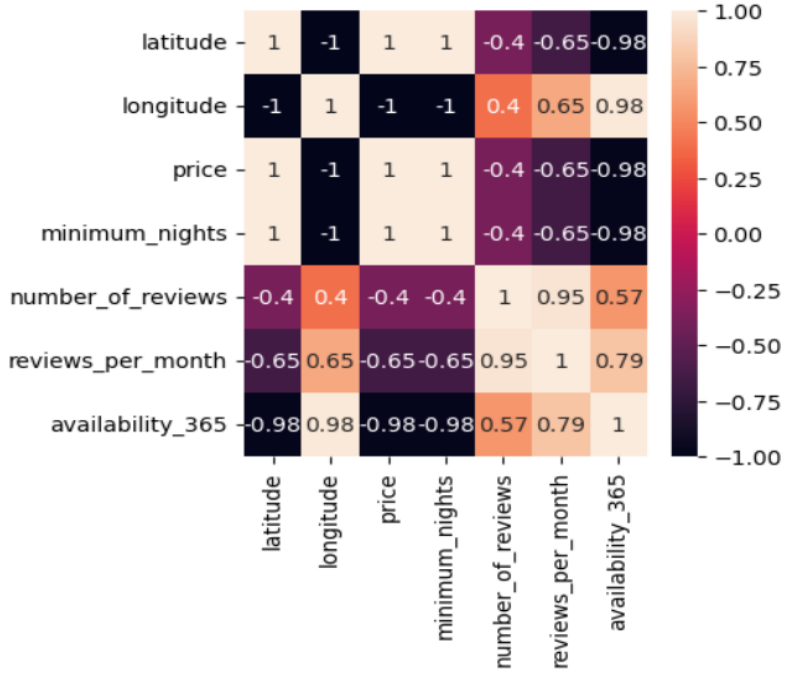
Geographical distribution of Airbnb listing

```
# Geographical Distribution of Airbnb Listings
plt.figure(figsize=(4,3))
sns.scatterplot(data = df,x='longitude',y='latitude',hue = 'room_type')
plt.title('Geographical Distribution')
plt.show()
```



Correlation between numerical columns

```
# Correlation between numerical columns
plt.figure(figsize=(4,4))
corr = data[['latitude','longitude','price','minimum_nights','number_of_reviews',
             |'reviews_per_month','availability_365']].corr()
corr
sns.heatmap(data=corr,annot=True)
plt.show()
```



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Key Insights

1. Price Trends:

- **Manhattan** has the most expensive listings due to central location, followed by Brooklyn has a competitive market.
- **Entire homes/apartments** cost significantly more than private or shared rooms.

2. Room Type Distribution:

- **Entire homes/apartments** are the most common, but **private rooms** offer budget-friendly options.It indicates customers pay more for privacy and comfort.

3. Outliers in Price:

- The presence of high-price outliers indicates a small luxury segment, while most customers prefer affordable listings, making the market largely price-sensitive.

4. Availability Patterns:

- Listings with **high availability** tend to have lower prices and more reviews, likely due to better guest experience.

5. Host Behavior:

- Some hosts manage **multiple listings**, indicating a trend toward professional hosting.

Recommendations

- **For Guests**:
  - Look for listings with high availability and good reviews for a better experience.
  - **Private rooms** in Brooklyn offer affordable stays compared to Manhattan.
- **For Hosts**:
  - Improve **availability** and **review response rates** to attract more bookings.
  - Manage pricing effectively to compete within the borough's market.

Conclusion

This project offers valuable insights into the New York Airbnb market, helping both guests and hosts make informed decisions. By using **EDA techniques**, I have identified key trends and developed actionable recommendations. Future improvements can involve advanced analytics and predictive modeling to further enhance the findings.

Thank you